

Does Mobility Difficulty Increase Fall-Related Fracture Risk?

Megha Radhakrishnan Sanitha

github.com/meggie2002

A matched-cohort analysis using public health survey data (NHANES 2017–2018)

The question: Are older adults who have trouble walking or moving around more likely to have broken a bone from a fall compared to similar patients (same age, same sex) without that trouble?

HOW I APPROACHED IT

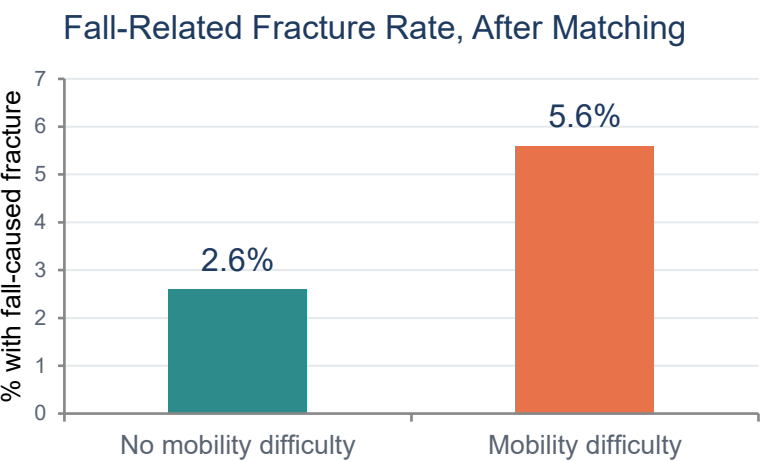
- 1 Combined three datasets**
linked demographics, fracture history, and mobility records for the same patients
- 2 Checked the data carefully**
verified missing answers were genuinely 'not applicable,' not just unanswered
- 3 Built a fair comparison**
paired each mobility-limited patient with a similar patient (same age, same sex) who wasn't
- 4 Tested the difference**
checked whether the gap in fracture rates was real, or just chance

WHAT I FOUND

Before comparing groups, I made sure they were actually similar in age and sex:

Before matching: **5.3-year age gap between groups**

After matching: **groups nearly identical (66.2 vs 66.1 yrs)**



KEY FINDING

2×

higher rate of fall-related fracture in patients with mobility difficulty

statistically significant
($p = 0.0006$)

Note: self-directed analysis using public NHANES survey data, applying a matched-cohort study design commonly used in health services research. Full analysis and code available on GitHub.